

RESEARCH ARTICLE

Elevating haptic interfaces: Dual-rate sampling and field programmable gate array implementation for multi-degree-of-freedom performance enhancement

Majid Koul¹  | Suhail Khosa² | Babar Ahmad²

¹Mechanical Engineering, Islamic University of Science and Technology, Kashmir, Jammu and Kashmir, India

²Mechanical Engineering, National Institute of Technology Srinagar, Jammu and Kashmir, India

Correspondence

Majid Koul, Mechanical Engineering, Islamic University of Science and Technology, Kashmir, India.

Email: majid.koul@iust.ac.in

Abstract

In this work, our primary focus centered on exploring the adaptability of the dual-rate sampling scheme proposed earlier to enhance the performance of multi-degree-of-freedom (multi-DOF) impedance-based haptic interfaces. The scheme employed independent sampling rates in a haptics controller, effectively mitigating the issue of reduced Z-width at higher sampling rates. A key aspect of our investigation was the intricate implementation of the dual-rate sampling scheme on a field programmable gate array (FPGA). This implementation on a logic hardware FPGA was challenging and led to the effective comparison of the uniform-rate and dual-rate sampling schemes of the multi-DOF haptic controller. We used an in-house developed two-DOF pantograph as the haptic interface and an FPGA for implementing the controller strategy. FPGA-based implementation presented challenges that were vital in testing controller performances at higher sampling rates. Virtual wall experiments were conducted to determine the stable and unstable interactions with the virtual wall. To complement the experimental results, we simulated the haptics force law for multi-DOF system on Simulink/MATLAB. Notably, the dual-rate sampling approach maintained the Z-width of the two-DOF haptic interface, even at higher controller sampling rates, distinguishing it from the conventional two-DOF uniform-rate control scheme. For example, employing a dual-rate sampling combination of 20–2 kHz consistently ensured the stable rendering of a maximum virtual stiffness of approximately 700 N/mm and maintained a reliable virtual damping range spanning from 0 to 5 Ns/mm. In contrast, the 20 kHz uniform-rate sampling approach failed to ensure interface stability in the presence of virtual damping, ultimately resulting in the unsuccessful implementation of any virtual stiffness at higher sampling rates. This work, therefore, establishes the potential of dual-rate sampling in the realm of haptic technology, with practical applications in multi-DOF systems.

KEYWORDS

control FPGA, haptic device, simulation virtual reality, Z-width

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1 | INTRODUCTION

Virtual reality and teleoperation systems augment human sensing and perception capabilities by enabling user interaction with simulated and remote objects. The presence of haptic feedback is essential in such systems for the user interactions to feel natural and immersive.^{1,2} More importantly when the intended application requires the emulation of object properties like stiffness, hardness, texture, and so on, that are perceptible only to touch.^{3–5} Haptic interfaces, which provide programmable touch feedback, thus form an indispensable part of these systems. These interfaces essentially act as a bidirectional link between the simulated virtual environment and the human operator. Tactile devices, kinesthetic devices, and exoskeletons exemplify the major categories of haptic interfaces in current use.⁶ Kinesthetic devices of the impedance type are, however, far more widespread than the other classes. Such interfaces are configured to sense the operator's input position, feed the same to the virtual environment model being rendered, determine the interaction forces/torques, and then apply the resultant force/torque via the device actuators at the end-effector.⁷ The potential applications of haptic interfaces have significantly expanded in recent years and are now spanned over fields that include medicine (surgical simulation and training, telemedicine, robotic surgery, rehabilitation), computer-aided design (three-dimensional [3D] modeling, virtual prototyping, assembling and disassembling of components), art and entertainment (virtual museums, gaming), scientific visualization (molecular simulation, mechanism dynamics), and micromanipulation, among others.^{6,8}

The degrees-of-freedom (DOFs), with regard to actuation, sensing, and force-feedback capabilities, is one of the important hardware specifications of haptic interfaces. Besides this, the number of DOFs that a haptic interface possesses is known to have an impact on the task performance.⁹ Single-DOF haptic interfaces have been widely used for research and specific application purposes owing to their simplified design, ease of hardware implementation, and lower costs.^{10–13} Most general-purpose applications, however, demand that the interfaces possess multiple DOF at sensing as well as actuation levels. For instance, the haptic interaction of a 3D avatar (virtual representation of the haptic interface's end-effector) with a 3D virtual object (object-object interaction) requires a six-DOF (three translational and three rotational) haptic interface.¹⁴ Similarly, in a point-object interaction paradigm wherein a point-type virtual avatar interacts with a 3D virtual object, a three-DOF haptic interface is required.¹⁵ Furthermore, a two-DOF haptic interface enables the rendering of constitutive properties like stiffness, damping, and so on of virtual objects in a planar workspace.^{16,17} A few examples of commercial multi-DOF haptic interfaces include the Omega series by Force dimension, Phantom series by Immersion, Falcon by Novint Technologies, and so on.

The *Z-width*, a measure of the dynamic range, refers to the spectrum of virtual impedances that a haptic interface can render stably. The haptic rendering of stiff objects such as bones in a virtual orthopedic surgery, a virtual tooth in a dental simulator, metal-metal

contact in a virtual mechanical assembly, and so on stipulates that the haptic interfaces should exhibit a broader *Z-width*.^{18,19} In effect, though, haptic interfaces, however, possess only a limited *Z-width*.²⁰ This limitation emanates as a result of the inherent device dynamics, notably damping and compliance, limited sensing and actuation capabilities that include sensor quantization and actuator saturation, and the digital implementation of virtual environments that entail sampling, time delay, filtering, and so on.^{20–22} The controller sampling rate and the device damping, in particular, have a marked effect on the *Z-width* of haptic devices.^{23,24} A higher device damping and faster sampling rates are known to translate into an enhanced *Z-width*.

The enhancement of the limited *Z-width* of haptic interfaces has been the subject of considerable research in the haptics community. Techniques that tend to increase the physical device damping are among the primary methods of improving the *Z-width*. A number of approaches relying on electrical, analog, digital, and passive means (like clutches and brakes) have been proposed to induce and augment damping into haptic interfaces.^{20,25–29} A major issue with such techniques, however, is that they degrade the device *transparency*—a critical requirement in haptics that nonrestrained motion should not feel impeded. Moreover, these techniques require additional hardware in the form of mechanical dampers, electronic components like resistors, capacitors and H-bridges, brakes, and so on.

Another line of approach involves the enhancement of *Z-width* by employing higher sampling rates.^{11,30} The issue of diminishing *Z-width* at elevated sampling rates as a consequence of estimating kinematic quantities such as velocity and acceleration from discrete position measurements, and the high-frequency noise induced thereof, however, limits their utility. Digital low-pass filtering redresses this issue to some extent but the filtering-induced time delay limits the level of filtering employed.^{31,32} Furthermore, the application of techniques like discrete-time adaptive windowing,³³ sliding mode control inspired Levant's differentiator,³⁴ and curve breaking algorithm³⁵ offer some alternate velocity estimation methods but they too cannot be employed at very high control rates, of the order of >10 kHz. Some very recently proposed methodologies to enhance haptic interface's *Z-width* can be found in the study by Chu et al.³⁶ and others.^{37–41} It is important to note that all of the above-cited techniques are based on the conventional *open-loop impedance-control* approach that utilizes a uniform sampling rate for the virtual environment rendering.⁴²

Multirate techniques have been effectively used in haptics for addressing issues related to *Z-width* enhancement, deformable object rendering, synchronous visual-haptic rendering, and so on.^{43–46} One such technique referred to as *dual-rate sampling* has been proposed by the authors elsewhere⁴⁴ to address the issue of diminishing the *Z-width* of haptic interfaces at higher sampling rates. The proposed scheme involves the decoupling of the stiffness and damping parts of the virtual environment into parallel feedback loops and rendering them at two distinct sampling rates. This control strategy thereby averts the instabilities arising due to poor velocity estimation from discrete position measurements, particularly, when lower-resolution sensors are employed. In subsequent works,^{47–49} the stability aspects

and limit cycle behavior of a single-DOF haptic interface under the dual-rate sampling scheme have been reported.

Most of the *Z-width* enhancement techniques, including the dual-rate sampling scheme, have been implemented on single-DOF haptic interfaces only. Despite the fact that most practical applications require multi-DOF interfaces, only a few techniques have attempted to enhance the *Z-width* of such interfaces.^{50,51} In this work, we accordingly intend to demonstrate the applicability of the dual-rate sampling scheme to multi-DOF haptic interfaces. The application of the dual-rate scheme to higher DOF haptic interfaces is interesting in two ways: first, it will potentially benefit the *Z-width* of multi-DOF haptic interfaces at higher sampling rates, second, it will establish the feasibility of the scheme to such systems. Toward this end, the dual-rate scheme is implemented here over a two-DOF haptic interface developed in-house. A key aspect of our investigation here is the intricate implementation of the dual-rate sampling scheme on a field programmable gate array (FPGA) at the digital level. This central focus requires addressing a range of complexities, including the development of digital logic for crucial components such as velocity estimation filters, collision detection mechanisms, and the incorporation of device kinematics for mapping the joint coordinates of the haptic interface with the operational space coordinates. Virtual wall experiments along with simulation of the multi-DOF haptics force law are demonstrated. Experimental and simulated *Z-width* plots for the haptic interface complement each other, demonstrating the advantages of the dual-rate scheme to multi-DOF systems. In addition, simulations demonstrate a vital role in studying the behavior of haptic interfaces before actually constructing them.

The remainder of this paper is organized as follows: Section 2 reviews the conventional and dual-rate control schemes with regard to the haptic rendering of virtual walls. In Section 3, the in-house developed two-DOF haptic device on which the dual-rate scheme is implemented is described and its kinematic analysis is reported. Section 4 exhaustively details the hardware implementation of the two-DOF dual-rate scheme on a logic device (an FPGA), while Section 5 illustrates the methodology for simulating the multi-DOF haptics force law. The experimental and simulation results are presented and discussed in Section 6, and finally, conclusions are listed in Section 7.

2 | REVIEW OF CONVENTIONAL AND DUAL-RATE HAPTIC RENDERING

In this section, we revisit the conventional and dual-rate haptic rendering techniques, as employed for the implementation of an archetypal virtual wall. The typical case of using optical encoders for position sensing and velocity estimation from discrete position samples by employing the finite difference method (FDM) is considered. At the outset, the architecture of a conventional uniform-rate haptics controller is described and its limitations at higher sampling rates are highlighted. Subsequently, the architecture of the dual-rate haptics controller is introduced and how it addresses the underscored limitations is deliberated.

In a typical uniform-rate haptics controller,⁴⁴ a virtual wall is akin to a constraint surface that provides both high impedance (during

contact) and displays null impedance (during phases of noncontact).¹¹ During interaction with a virtual wall, a haptic device provides feedback forces in proportion to the wall impedance, which is typically a function of the stiffness K and damping B of the wall.

In the usual implementation, the sampled position data represented by θ_d^+ acquired from device encoders is digitally differentiated using the backward difference method for estimating the device velocity, given as $\dot{\theta}_d^+(z) = \frac{(z-1)}{Tz} \theta_d^+(z)$, with z and T being the discrete-time variable and the uniform controller sampling period, respectively. The uniformly sampled position and velocity data are then subsequently utilized for generating necessary feedback force/torque when the user penetrates the wall during exploration.

For a rotational type one-DOF haptic device, the feedback torque, represented by τ_{VE} , resulting from its motion within or outside the virtual wall is governed by

$$\begin{cases} \tau_{VE} = K\Delta\theta + B\omega & : \text{Within the virtual wall,} \\ = 0 & : \text{Outside the virtual wall,} \end{cases} \quad (1)$$

where $\Delta\theta$ and ω are the angular penetration and angular velocity of the one-DOF haptic device within the virtual wall, respectively. Equation 1, also known as the haptics force law, represents a penalty-based method for simulating virtual wall interaction using a one-DOF haptic device. These models are extensively employed in haptics applications due to their ease of modeling and computational simplicity.^{10,22}

In this seemingly simple control scheme, factors like the sampling operation, quantization, differentiation-induced noise, asynchronous wall switching, and so on tend to make an otherwise passive virtual wall exhibit active behavior by inducing nonphysical energy into the system.²¹ The generated energy, if not dissipated, by the intrinsic device damping or through proper control can cause limit cycles and in the worst cases lead to device instability and hence limit the interface *Z-width*.^{11,20–22} The sampling operation, in particular, is known to be the major contributor to energy generation and quantization is known to exacerbate its effect. The situation tends to worsen at higher sampling rates owing to the loss of information due to discrete sampling coupled with the introduction of high-frequency noise due to backward difference-based velocity estimation. Although aggressive low-pass filtering helps in attenuating the noise in the estimated velocity, the inherent time delay induced by filtering adds to the instability of the system.^{20,31,32} This restricts the utility of the conventional uniform-rate sampling scheme to lower sampling rates and thereby to soft model rendering (smaller virtual stiffness and narrower *Z-width*) only.

Alternatively, in the one-DOF dual-rate haptics controller as proposed, implemented, and analyzed by Koul and colleagues,^{44,47,49} the rendering of the virtual wall impedance parameters is decoupled into two parallel feedback loops, referred to as the position loop and velocity loop, respectively (Figure 2 of Koul et al.⁴⁴). In other words, instead of rendering the wall impedance at a uniform higher sampling rate ($f = 1/T$), its constituents that include the stiffness K and damping B are rendered independently at two distinct sampling rates ($f_1 = N/T$, $f_2 = 1/T$, $N \in \mathbb{Z}$) under the dual-rate sampling paradigm. Since higher sampling rates exacerbate the velocity estimation, the damping part B , in the velocity loop, is rendered at relatively lower sampling rates,

at $f_2 = 1/T$, while the stiffness part K , in the position loop, is simultaneously rendered at elevated sampling rates, $f_1 = N/T$. The proposition is based on the fact that increasing the position loop's sampling frequency enhances the stable range of virtual stiffness K while reducing the sampling frequency of the velocity loop prevents velocity exacerbation.^{10,11,20} This decoupled rendering scheme has been previously demonstrated to enable the emulation of higher wall stiffness and likewise a broader Z -width than those admitted by the conventional uniform-rate scheme for single-DOF haptic interfaces.⁴⁴ Figure 1 depicts the block diagram of a dual-rate haptics controller as employed in a generic multi-DOF haptic interface. The details pertaining to the hardware implementation of this extended multi-DOF dual-rate sampling scheme are provided in Section 4.

3 | DEVICE DESCRIPTION AND KINEMATICS

Haptic interfaces with parallel kinematic architecture are known to possess high stiffness, low inertia, good positioning resolution, and so on, which are among the desirable requirements for achieving a broader Z -width.^{16,52} In view of this, a five-bar parallel type haptic interface was selected here that simulates forces within a planar rectangular workspace. The fabricated device is depicted in Figure 2A along with its kinematic diagram. The device functions as a congruent pantograph mechanism with a base length of zero. The remaining link lengths were uniformly set to 10 cm each. This specific design choice resulted in an expanded usable workspace for the device compared to those with a nonzero base length. As a result, the device may resemble a four-bar mechanism rather than a five-bar pantograph mechanism. However, the key distinction lies in the

fact that a four-bar mechanism is a single-DOF device, with only one of its links being both actuated. Conversely, in the congruent design pantograph, two of its links are actuated. Additionally, in a four-bar mechanism, only three links are movable, while the fourth link is grounded, whereas in a congruent design pantograph mechanism, four of the links are movable, while the fifth is grounded. The selection of the link lengths was driven by a reasonable intuition while taking into account the need for an adequate workspace and minimal force output for the experiments. Note that this two-DOF haptic interface is a congruent joint pantograph mechanism with torque enhancers, whereas the one reported by Ramstein and Hayward¹⁶ is a direct drive setup with nonzero base length and no torque enhancers, thereby requiring higher torque motors for similar end-effector force output.

The device was actuated by two brushless DC motors (Make—Maxon RE-25) with integrated 1000 counts per revolution (CPR) encoders that were mounted at the device base. The position resolution was amplified to 4000 due to quadrature decoding of the encoder signals and further to 40 000 owing to the torque enhancers that were conjoined with the active links and provided a 10-fold gear ratio. The device was fabricated using rapid prototyping with Polyamide PA 2200, which possesses a yield strength of 45 MPa and Young's Modulus of 1700 MPa. Through selective laser sintering technology, it was ensured that the device had sufficient structural stiffness for conducting virtual wall experiments. Furthermore, fine journal bearings were installed in the joints to reduce friction. Additionally, a capstan drive and a spring were utilized to enhance torque and pretension, respectively. Further details pertaining to design iterations and the device synthesis can be found in Koul et al.⁵³

Haptic interfaces featuring two actuated DOF represent a simple example of multi-DOF systems characterized by a nonlinear relationship

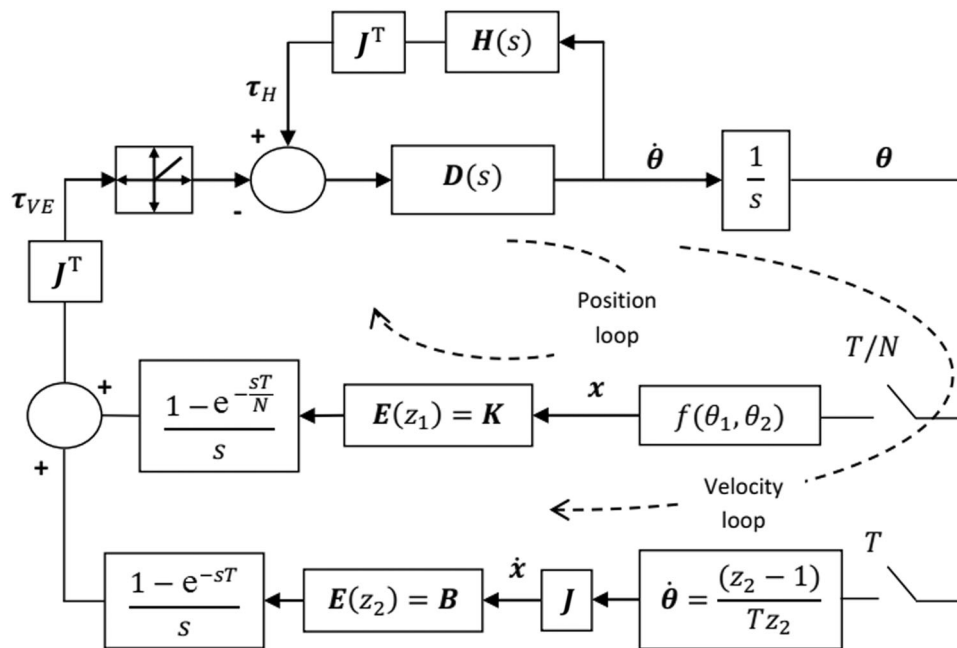


FIGURE 1 Block diagram representation of multi-degree-of-freedom dual-rate haptic rendering.

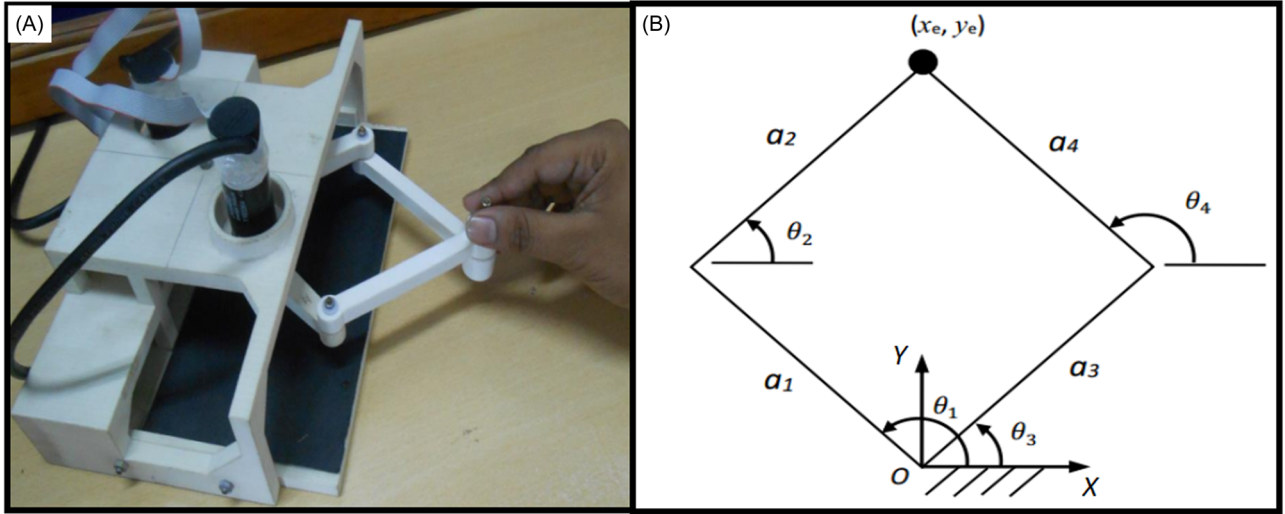


FIGURE 2 Actual two degrees-of-freedom (DOFs) pantograph-type haptic device and its kinematic diagram. (A) Fabricated two-DOF haptic device being manipulated by a user, a manipulandum; (B) kinematic diagram of the device.

between the end-effector coordinates and joint angles. With reference to the kinematic diagram in Figure 2B, the ordered pair (x_e, y_e) denotes the end-effector position in the operational space, considered the X-Y plane, measured with respect to the origin O or the fixed end. The usable or singularity-free workspace dimensions were 17 cm along the X axis and 7 cm along the Y axis. The choice of this workspace was limited due to the presence of singularities within the workspace as well as boundary singularities that are typical of parallel manipulators. Along the Y axis (length-wise), the workspace spanned from 10 to 17 cm from the origin O, whereas along the X axis (width-wise), the motors were positioned at the middle of the workspace, creating a distance of 8.5 cm from either side of the origin. The available workspace provided sufficient room for conducting the virtual wall experiments as well as for simulating various two-dimensional virtual environments. The rectangular usable workspace of the device was determined based on the following geometric constraints:

$$\begin{aligned} (x_e - a \cos \theta_1)^2 + (y_e - a \sin \theta_1)^2 &= a^2, \\ (x_e - a \cos \theta_3)^2 + (y_e - a \sin \theta_3)^2 &= a^2, \end{aligned} \quad (2)$$

where a denotes the uniform link length, while θ_1 and θ_3 are the active joint variables. By simplifying Equation 2, the forward kinematic map can be expressed as a straightforward nonlinear function of the active joint angles as

$$\begin{aligned} x_e &= 2a \cos\left(\frac{\theta_1 - \theta_3}{2}\right) \cos\left(\frac{\theta_1 + \theta_3}{2}\right), \\ y_e &= 2a \cos\left(\frac{\theta_1 - \theta_3}{2}\right) \sin\left(\frac{\theta_1 + \theta_3}{2}\right). \end{aligned} \quad (3)$$

Further simplifications of the above equation yield

$$\begin{aligned} x_e &= a(\cos \theta_1 + \cos \theta_3), \\ y_e &= a(\sin \theta_1 + \sin \theta_3). \end{aligned} \quad (4)$$

These forward kinematic equations are extremely important for virtual wall rendering applications wherein the forces of interaction are invariably estimated in the operational space coordinates, while the position sensors are localized at the active joints. The inversion of Equation 4 can be used to obtain the inverse kinematics of the two-DOF device. In doing so, the following expressions for the active joint angles are obtained:

$$\begin{aligned} \theta_1 &= \frac{1}{2} \left[2 \cot^{-1} \left(\frac{x_e}{y_e} \right) + \cos^{-1} \left(\frac{x_e^2 + y_e^2 - 2a^2}{2a^2} \right) \right], \\ \theta_3 &= \frac{1}{2} \left[2 \cot^{-1} \left(\frac{x_e}{y_e} \right) - \cos^{-1} \left(\frac{x_e^2 + y_e^2 - 2a^2}{2a^2} \right) \right]. \end{aligned} \quad (5)$$

The Jacobian matrix, $J(\Theta)$, which relates the active joint velocities $\dot{\theta}_1$ and $\dot{\theta}_3$ to the end-effector velocities $\dot{x}_e$ and $\dot{y}_e$, as well as the joint torques to the end-effector forces can be obtained by taking the time-derivative of the forward kinematic equations given by Equation 4. The end-effector velocities $\dot{x}_e$ and $\dot{y}_e$ are accordingly given by

$$\begin{aligned} \dot{x}_e &= -a(\dot{\theta}_1 \sin \theta_1 + \dot{\theta}_3 \sin \theta_3), \\ \dot{y}_e &= a(\dot{\theta}_1 \cos \theta_1 + \dot{\theta}_3 \cos \theta_3). \end{aligned} \quad (6)$$

After rearranging the above equations in the form, $\dot{\mathbf{X}} = J(\Theta)\dot{\Theta}$, with $\dot{\mathbf{X}} = [\dot{x}_e, \dot{y}_e]^T$, and $\dot{\Theta} = [\dot{\theta}_1, \dot{\theta}_3]^T$, being the vectors of end-effector velocities and active joint velocities, respectively, the Jacobian matrix, $J(\Theta)$, turns out to be

$$J(\Theta) = a \begin{bmatrix} -\sin \theta_1 & -\sin \theta_3 \\ \cos \theta_1 & \cos \theta_3 \end{bmatrix}. \quad (7)$$

It is a well-known fact that multi-DOF robots and likewise haptic interfaces lack uniform force rendering capability in their workspace owing to the nonlinear relation between operational space and joint

space coordinates. The relation between the end-effector force vector and the corresponding joint torque is given by the following relation:

$$\mathbf{F} = \mathbf{J}^T(\Theta)\boldsymbol{\tau}, \quad (8)$$

where $\mathbf{F} = [f_x, f_y]^T$ and $\boldsymbol{\tau} = [\tau_1, \tau_2]^T$, where f_x and f_y represent the forces applied by the end-effector along the x and y directions in the operational space, and τ_1 and τ_2 represent the corresponding torque applied at the active joints. The current two-DOF haptic interface achieved a minimum force of 30 N at the extreme ends of the usable workspace, as illustrated in Figure 3B, utilizing the stall torque (maximum torque allowed for a brief duration) of the DC motors used in this work. A recent study by Łacki et al.⁵⁴ has revealed the significant impact of the base link length on the force rendering capability of five-bar parallel haptic interfaces. Furthermore, the congruent design with equal link lengths and zero base link length, similar to haptic interfaces under consideration, has been reported to generate the widest range of forces.

Furthermore, manipulability being another kinematic feature in the context of robotics and mechanical systems refers to the ability of a mechanism to control the movement and orientation of its end-effector within its workspace.⁵⁵ It quantifies how well the mechanism can reach and maneuver in different directions and orientations. Mathematically, manipulability measure at a particular state vector, Θ , is a scalar value represented as “ w ” and derived from the Jacobian matrix of the mechanism as

$$w = \sqrt{\det(\mathbf{J}(\theta)\mathbf{J}^T(\theta))}. \quad (9)$$

Manipulability contours in Figure 3A are vital tools in haptic device design, offering visual representations of the reachable workspace and providing insights into device performance. These

contours visualize regions within the workspace where users can effectively manipulate objects and interact with virtual environments, guiding designers in optimizing device coverage and usability. Additionally, manipulability contours aid in evaluating device dexterity, agility, and efficiency, facilitating the development and optimization of control algorithms for accurate force and motion rendering.

Yet another significant feature in haptics device design, referred to as force manipulability ellipsoid, is essentially a mathematical representation to analyze the capability of a manipulator to generate forces and/or torques (together referred to as wrench) in different directions and orientations within its operational workspace. Using Equation (8), we can obtain the following expression:

$$\boldsymbol{\tau}^T \boldsymbol{\tau} = \mathbf{F}^T \mathbf{J} \mathbf{J}^T \mathbf{F}. \quad (10)$$

For all joint torques to be in a unit hypersphere, we have $\boldsymbol{\tau}^T \boldsymbol{\tau} = 1$, which implies

$$\mathbf{F}^T \mathbf{J} \mathbf{J}^T \mathbf{F} = 1. \quad (11)$$

Since the matrix $\mathbf{J} \mathbf{J}^T$ is a real and symmetric matrix, the singular value decomposition of the same would lead to

$$\mathbf{J} \mathbf{J}^T = \mathbf{S} \boldsymbol{\Lambda} \mathbf{S}^T, \quad (12)$$

where the diagonal matrix $\boldsymbol{\Lambda}$ contains the eigenvalues on the diagonals, and the matrix $\mathbf{S}$ contains the eigenvectors on its columns. These eigenvalues and eigenvectors correspond to the dimensions of the ellipsoid and the coordinate directions of the ellipse at a particular state. The force manipulability ellipsoids at various state vectors Θ are plotted in Figure 3B. As discussed in Yoshikawa,⁵⁵ the above results confirm that the direction in which a large manipulating force can be generated is the one in which the manipulability is poor and vice versa.

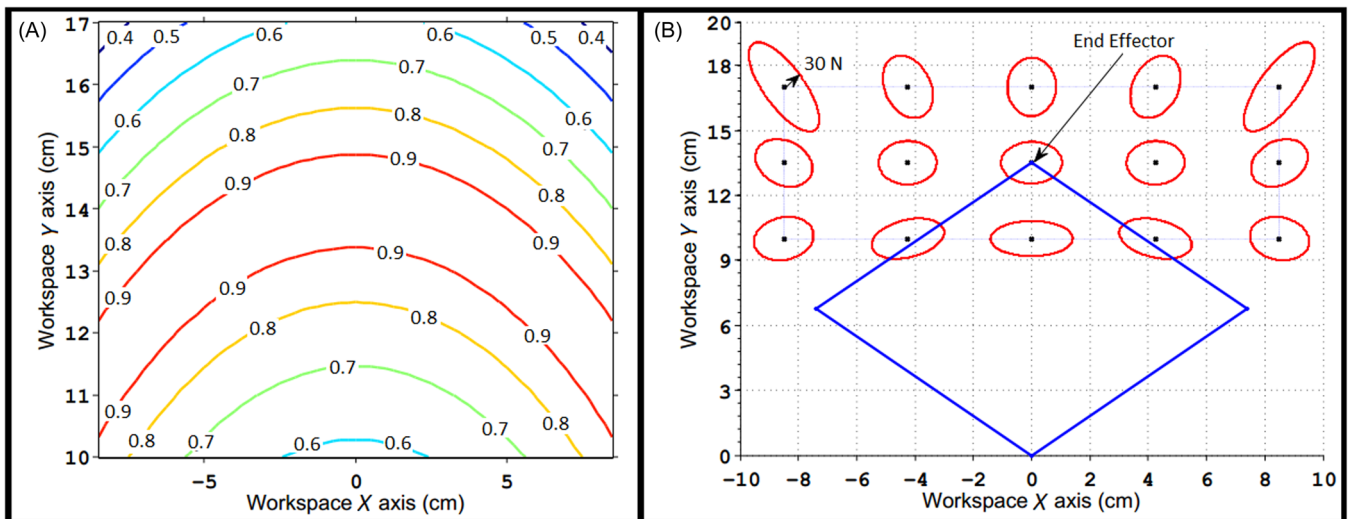


FIGURE 3 Design features of the two degrees-of-freedom haptic device used in this study. (A) manipulability contours; (B) force manipulability ellipsoids.

4 | MULTI-DOF DUAL-RATE CONTROL IMPLEMENTATION

This section details the hardware implementation of the dual-rate control approach for the two-DOF interface described in the previous section. It is important to note that while the two-DOF interface is the simplest of the multi-DOF interfaces its control implementation entails all the important elements that other generic multi-DOF interfaces do. For instance, the realization of the two-DOF haptic rendering requires the implementation of the device kinematics that map the joint-space coordinates to the operational-space coordinates, the force-torque mapping via the device Jacobian, and so on, apart from the control law implementation. The same is true for other generic multi-DOF devices. Hence, the methodology presented here can be well extended to other higher DOF devices as well.

In view of our interest, at higher sampling rates, of the order of 20 kHz, the control strategy was implemented on a FPGA—Altera Cyclone®-II. This choice was necessitated as ordinary controller platforms like Arduinos are incapable of effectively sustaining such higher sampling rates as they tend to experience information loss on account of missed encounter counts and so on.⁵⁶ Figure 4 depicts some of the modules that need to be designed on an FPGA for a generic multi-DOF haptic force law implementation.

At each position within the operational space, the device can be considered to render two virtual walls instantaneously: one in the horizontal X-direction and the other one in the vertical Y-direction. The forces rendered at the end-effector can thus be computed as

$$\begin{Bmatrix} f_x \\ f_y \end{Bmatrix} = \begin{bmatrix} K_x & 0 \\ 0 & K_y \end{bmatrix} \begin{Bmatrix} \Delta x \\ \Delta y \end{Bmatrix} + \begin{bmatrix} B_x & 0 \\ 0 & B_y \end{bmatrix} \begin{Bmatrix} \dot{x} \\ \dot{y} \end{Bmatrix}, \quad (13)$$

where K_x , B_x , and Δx are the virtual stiffness, damping, and the wall penetration in the X-direction and likewise for the Y-direction, respectively. The digital implementation of this two-DOF force law required that the involved quantities K_x , B_x , and so on be represented by a certain bit value in the FPGA. The minimum bit value for each of these quantities was determined by considering factors such as the available

position resolution, stall torque of the motors, and so on, as will be detailed in the later part of this section. The foremost step in the implementation is the quadrature decoding of the encoder signals for determining the instantaneous active joint positions. A 12-bit counter (maximum value 4096) that maintains the joint position in terms of the encoder counts was used for the purpose. The presence of transcendental functions such as $\sin\theta$ and $\cos\theta$ in the forward kinematic equations and the Jacobian matrix (see Equations 3, 4, 6, and 7) necessitated the conversion of the active joint motion information available in encoder counts into radians. This required determining the relationship between the minimum position resolution in terms of encoder counts and radians. As already mentioned, the 1000 CPR encoder resolution gets magnified to 40 000 due to quadrature decoding and the gear ratio of torque enhancers. The minimum displacement at the joints is thus given by

$$40\,000 \text{ CPR} = 2\pi \text{ radians},$$

$$1(\text{encoder count}) = \frac{2\pi}{40\,000} \text{ radians}. \quad (14)$$

The position count obtained from the counter module was thus converted into angular motion in radians, as per Equation 14. This angular motion is typically a fractional quantity and is thereby expressed in the standard logic vector (*std_logic_vector*) format on an FPGA. In addition, the coefficients of the low-pass filter employed in conjunction with FDM for velocity estimation are also usually expressed as fractional quantities and accordingly in the *std_logic_vector* format. To implement these fractional quantities, it is more convenient to use the fixed-point format instead of the floating-point format.⁵⁷ Consequently, a fixed-point converter module was employed to convert these fractional values from the *std_logic_vector* form to the fixed-point form.

The next step involved continuously computing the end-effector position in the operational space as expressed via the forward kinematics equations given by Equations 3 and 4. We employed a COordinate Rotation DIgital Computer (CORDIC) algorithm⁵⁸ to compute the trigonometric functions involved in these forward kinematics equations and the Jacobian matrix given by Equation 7.

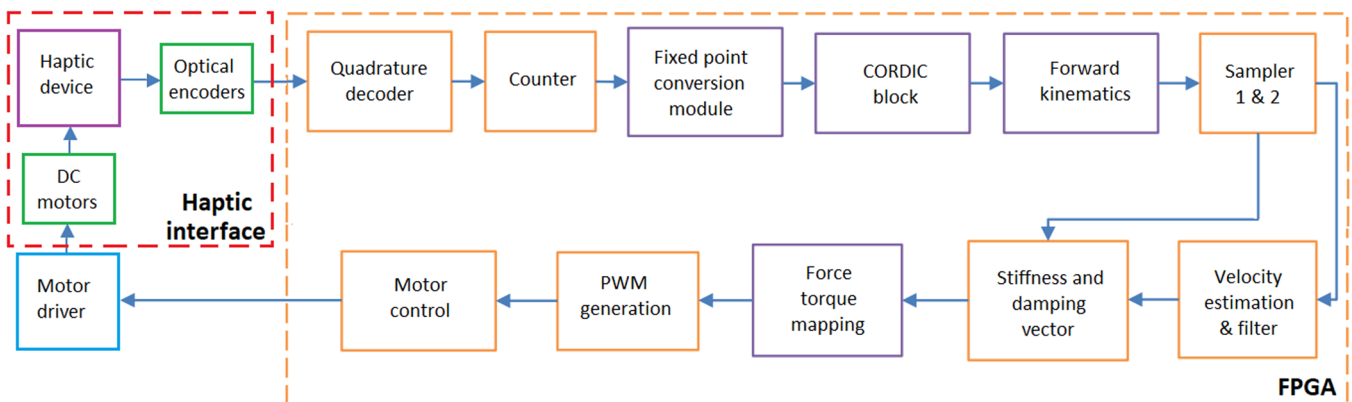


FIGURE 4 Components designed in field programmable gate array (FPGA) for the implementation of force law for a multi-degree-of-freedom haptic interface.

CORDIC is an iterative shift-and-add technique commonly used in embedded systems, such as calculators, DSPs, and supercomputers, with a particular focus on robotics applications.⁵⁹ Specifically, the rotation mode of the CORDIC algorithm proposed by Vladimirova and Tiggele⁶⁰ was utilized for *sine* and *cosine* computation in this work.

The next step involved the determination of the end-effector velocity components $\dot{x}_e$ and $\dot{y}_e$ that are given by Equation 6 and are related to the joint velocities $\dot{\theta}_1$ and $\dot{\theta}_3$ via the Jacobian matrix in Equation 7. It is to be recalled that in dual-rate sampling implementation the joint velocities $\dot{\theta}_1$ and $\dot{\theta}_3$ and hence the end-effector velocity components $\dot{x}_e$ and $\dot{y}_e$ were computed with a lower sampling frequency of $f_2 = 1/T$, whereas the joint position data $\dot{\theta}_1$ and $\dot{\theta}_3$ and hence the end-effector position components $\dot{x}_e$ and $\dot{y}_e$ were computed at much higher sampling rates of $f_1 = N/T$. Furthermore, it is important to note that here the evaluation of the end-effector velocity components needs additional multiplications and additions and accordingly more digital resources while no such operations are needed for the single-DOF case.⁴⁴ This complexity increases in proportion to the device's DOF and therefore there has been a significant focus on devising efficient approaches to implement real-time control systems in embedded systems such as FPGA's.

The feedback forces resulting from the virtual wall interaction were computed using Equation 13 for which the desired virtual stiffness and damping coefficients were set in the controller itself. The depth of penetration required for the virtual stiffness implementation was evaluated as

$$\begin{aligned}\Delta x &= x_{wp} - x_{cp}, \\ \Delta y &= y_{wp} - y_{cp},\end{aligned}\quad (15)$$

where the subscripts wp and cp refer to the wall position and current device position, respectively. The units of position coordinates were taken care of in the program. For instance, if the current position was calculated in tenths of a millimeter, the same units were used for specifying the wall position. Since the maximum force that the device can apply depends on its position in the workspace, as demonstrated by the Force Manipulability Ellipsoids shown in Figure 3, the maximum forces at a specific position were determined by utilizing the stall torque of the motors through the force-torque mapping equation, $F = J^{-T}\tau$, as

$$\begin{Bmatrix} f_x^{\max} \\ f_y^{\max} \end{Bmatrix} = J^{-T} \begin{Bmatrix} \tau_1^{\text{stall}} \\ \tau_2^{\text{stall}} \end{Bmatrix}.\quad (16)$$

The maximum virtual stiffness and damping coefficients were then estimated as

$$K_x^{\max} = \frac{f_x^{\max}}{\Delta x} \text{ and } B_x^{\max} = \frac{f_x^{\max}}{\dot{x}}.$$

Similarly,

$$K_y^{\max} = \frac{f_y^{\max}}{\Delta y} \text{ and } B_y^{\max} = \frac{f_y^{\max}}{\dot{y}}.\quad (17)$$

For the experiments, a simple virtual wall (located in the negative Y-direction) was selected here for implementation, as is schematically shown in Figure 8. With this choice, the only nonzero virtual wall coefficients were K_y and B_y . Additionally, the virtual wall position was set as $y = 145$ mm from the origin, allowing for flexibility in the experiments with x ranging from 50 mm on either side of the origin.

From the force-torque mapping, with $y = 145$ mm, we have

$$f_y^{\max} = K_y(\text{max bits}) \times 1\text{bit}(\Delta y) = \frac{\tau(\text{max bits})}{688.75},$$

$$K_y(\text{max bits}) = \frac{4096}{688.75} = 5.947,\quad (18)$$

where Δy represents the minimum change in the Y-direction position value, which is implemented as 1 bit and is equivalent to 0.1 mm in the FPGA. Correspondingly, the maximum force in the Y-direction is 43.41 N. Hence,

$$\begin{aligned}f_y^{\max} &= K_y(\text{max}) \times \Delta y, \\ 43.41 \text{ N} &= K_y(1 \text{ bit}) \times 5.947 \times 0.1 \text{ mm}, \\ K_y(1 \text{ bit}) &= 72.9 \text{ N/mm}.\end{aligned}\quad (19)$$

Similarly, for virtual damping implementation, the end-effector's minimum velocity is given by

$$\begin{Bmatrix} \dot{x}_{\min} \\ \dot{y}_{\min} \end{Bmatrix} = J^T \begin{Bmatrix} \dot{\theta}_{1,\min} \\ \dot{\theta}_{3,\min} \end{Bmatrix},\quad (20)$$

where the minimum active joint velocities are $\dot{\theta}_{1,\min}$, $\dot{\theta}_{3,\min} = \frac{2\pi}{40000}$ rad/s.

In accordance with Equation 7, and for $y = 145$ mm, the minimum end-effector velocity in the Y-directions is equal to $\dot{y}_{\min} = 21.6377$ mm/s. Hence, we have

$$\begin{aligned}f_y^{\max} &= B_y(\text{max}) \times \dot{y}_{\min}, \\ 43.41 \text{ N} &= B_y(1 \text{ bit}) \times 5.947 \times 21.6377 \text{ mm}, \\ B_y(1 \text{ bit}) &= 0.3374 \text{ Ns/mm}.\end{aligned}\quad (21)$$

A force-torque mapping block was employed to calculate the necessary active joint torques depending on the desired feedback forces at the end-effector by using

$$\tau = J^T F,\quad (22)$$

where $\tau = [\tau_1, \tau_3]^T$ is the active joint torque vector, while $F = [f_x, f_y]^T$ represents the vector of feedback-forces at the end-effector. It is to be noted that explicit expressions for the joint torques were used to perform the multiplication process with the Jacobian matrix on the FPGA, taking into account input feedback forces from the virtual environment. The calculated torque for the Y-direction, as determined by Equation 22, was subsequently used to generate pulse-width modulation (PWM) signals via a comparator logic module. The module featured a 12-bit running counter that produced a sawtooth wave whose values were constantly compared to the values of $K_y\Delta y + B_y \dot{y}$ resulting in the

generation of a logic “1” or “0” as a PWM signal. The motor control module then commanded the PWM signal, along with the rotation direction, to the appropriate H-bridges/motor drivers. A custom-developed motor controller board that housed the H-bridge circuits and line receivers for denoising encoder signals was used for the current control of motors. The experimental results are provided in Section 6.

5 | MULTI-DOF HAPTICS FORCE LAW SIMULATION

The theoretical and simulation-based analyses of the multi-DOF dual-rate control scheme are helpful in identifying the limits of the control scheme, specifically the *Z-width*, and also facilitate in evaluating the combined effect of factors like quantization, time delay, and the dual-rate sampling approach on their performance.⁴⁷ A theoretical analysis however requires the knowledge of the coupled dynamics of the multi-DOF device, the virtual environment, the human operator, and the controller, which is difficult to obtain.⁵² On the other hand, commercial software packages such as Simulink/SimMechanics from MATLAB act as essential tools for modeling, simulation, and control validation of haptics and robotics systems.

In this section, we intend to demonstrate how the multi-DOF dual-rate haptic control scheme can be simulated using such commercial software packages. Toward this end, a realistic model of the two-DOF dual-rate haptic control causality is modeled and simulated in Simulink/SimMechanics and the stability boundaries are obtained for various sampling rate combinations. Stability analysis of haptic interfaces is vital due to the direct physical contact between the device and the operator.^{23,47} The simulation methodology is briefly elaborated here and further details can be found in Koul et al.⁶¹

Realistic simulation of the two-DOF haptics force law requires modeling of all the components of the haptics controller depicted in Figure 1. To simplify the underlying modeling, the two-DOF dual-rate haptics controller is subdivided into the following components: mechanical (haptic device structure, joint damping, transmission), electrical (DC motor, current amplifier), and electronics (encoder, controller, H-bridge). The mechanical modeling relies on the knowledge of the device structure and its kinematics and dynamics. The electrical aspect requires establishing the relationship between applied potential and torque generated, while the electronic part involves defining the input-output relations between the components. In what follows, these subcomponents of two-DOF dual-rate haptics controller are modeled under the following two headings: (1) modeling haptic device and (2) modeling haptics controller.

5.1 | Modeling haptic device

For modeling the two-DOF haptic device, a stick model was developed, necessitating prior knowledge of the device's geometric and inertial parameters. The identification procedure proposed in Abbott and Okamura¹¹ was modified here to determine the inertial parameters including link mass, joint damping, and so on of the device under

consideration. The model was developed by utilizing the basic building blocks available in the SimMechanics library. Specifically, the blocks used included the link modules, sensor ports, and actuator ports. The link modules are responsible for defining each link's properties, such as its coordinates in either the global or local frame. Additionally, the link module specifies the inertia matrix, the mass of the link, and the center of mass. A joint sensor block was utilized to measure joint variables such as position, velocity, and reaction torque at desired joints. Similarly, the actuator block allows for the application of torque or force at the desired joints or different locations within the device. Apart from this direct device modeling, SimMechanics also provides the feature to import the device model based on existing computer aided design (CAD) files. This enables the simulation of the device's precise shape, geometry, and inertial properties. Figure 5A depicts the developed stick model as displayed in the SimMechanics animation window, while Figure 5B shows the imported CAD model of the device.

5.1.1 | DC motor and joint friction model

Since the actuation in the actual device is achieved by using DC motors and friction exists in the device joints, a realistic simulation of the device necessitated the incorporation of the dynamic model of the DC motor and the joint friction model. The DC motor model expresses the relationship between the motor current and applied voltage in terms of the motor parameters and back EMF, whereas the joint friction model considers the combined effect of viscous and Coulomb damping.

The torque, τ_m , generated by the DC motor is directly proportional to the electric current, i_m , passing through it. This relationship can be expressed using the motor torque constant, K_m , as

$$\tau_m = K_m i_m. \quad (23)$$

Furthermore, as an electrical system, the input voltage, V , to the DC motor is related to the current, i_m , according to

$$L \frac{di_m}{dt} + i_m R + \frac{1}{C} \int i_m dt = V - K_e N_m \dot{\theta}, \quad (24)$$

where K_e , L , C , and R denote the back EMF constant, inductance, capacitance, and resistance of the motor coil, respectively. If the motor capacitance, C , is assumed to be negligible, the above equation reduces to

$$i_m(s) = \frac{V - K_e N_m \dot{\theta}}{Ls + R}. \quad (25)$$

Accordingly, the actuator torque, τ_m , is related to the input voltage, V , as

$$\tau_m = \frac{N_m K_m (V - K_e N_m \dot{\theta})}{Ls + R}. \quad (26)$$

The negative sign of the term $K_e N_m \dot{\theta}$ signifies that in effect the back EMF of the DC motor functions as a frictional torque in

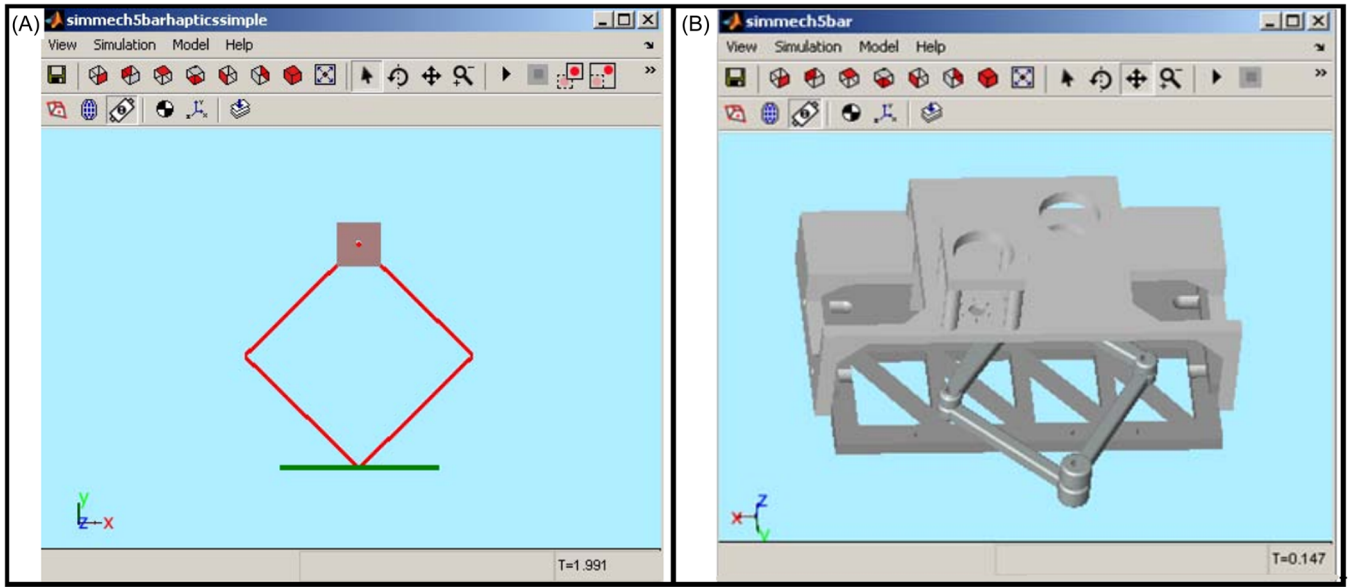


FIGURE 5 Two degrees-of-freedom (DOF) haptic interface models in SimMechanics environment of Simulink/MATLAB. (A) Basic stick model of the two-DOF pantograph mechanism; (B) imported CAD model of the developed pantograph from CAD Software. CAD, computer aided design.

proportion to the motor's angular velocity. This frictional torque plays a substantial role in haptics applications. Consequently, the effective viscous coefficient at the active joints of the haptic device is

$$b = b_m + b_e, \text{ where } b_e = \frac{N_m^2 K_m K_e \dot{\theta}}{L_s + R}. \quad (27)$$

Furthermore, a viscous and Coulomb friction (viscous damping coefficient, b_m , and Coulomb friction coefficient, f_c) model was defined for active joints with the already estimated coefficients. The implementation of Coulomb friction involved utilizing the signum function, $\text{sign}()$, which assigns a positive or negative sign depending on the joint velocity direction. The frictional torque, τ_f , acting at the active joints is thus governed by the equation

$$\tau_f = b_m \dot{\theta} + f_c \text{sign}(\dot{\theta}), \text{ where } \text{sign}(\dot{\theta}) = \begin{cases} +1; & \text{for } \dot{\theta} > 0, \\ -1; & \text{for } \dot{\theta} < 0. \end{cases} \quad (28)$$

Note that the frictional torque, τ_f , was implemented using the velocity obtained directly from the joint sensor module, which provided delayless and noise-free velocity. This approach was chosen to incorporate the natural frictional effect that occurs in real physical systems, as opposed to using a discrete method such as virtual damping in a simulated environment. Moreover, both active joints were assumed to have similar friction coefficients, while the passive joints were considered frictionless.

5.2 | Modeling haptics controller

Simulink allows the modeling of almost every component of a generic multi-DOF haptic control system. Figure 6 depicts the components of

the controller modeled in Simulink, which simulates the two-DOF haptics force law as given in Equation 13. For ease of modeling, the control scheme was implemented using the following modules.

5.2.1 | Forward kinematics module

At each simulation time step, the active joint positions in radians, acquired from the joint sensor modules, were first passed through the quantization blocks that enabled to mimic the effect of encoder resolution. The quantized angular positions were thereafter fed to the *Forward Kinematics* block wherein the devices' forward kinematic equations given by Equation 4 were analytically preprogrammed to determine the end-effector's operational space coordinates.

5.2.2 | Velocity estimation module

The end-effector velocity components were determined in accordance with Equation 6, which required the premultiplication of the joint velocities by the Jacobian matrix given by Equation 7. A *Zero-order Hold* block was used to sample the quantized joint positions at desired sampling rates. A *Joint Rate Evaluation* module was developed wherein the sampled and quantized active joint positions were digitally differentiated using the *discrete derivative* block, subsequently low-pass filtered, and then multiplied by the Jacobian matrix resulting in the end-effector's velocities. A second-order Butterworth filter, designed using the *fdatool* tool in MATLAB®, was used for filtering the estimated joint velocities. The filtering operation and Jacobian matrix were implemented using preprogrammed MATLAB® function blocks available in Simulink.

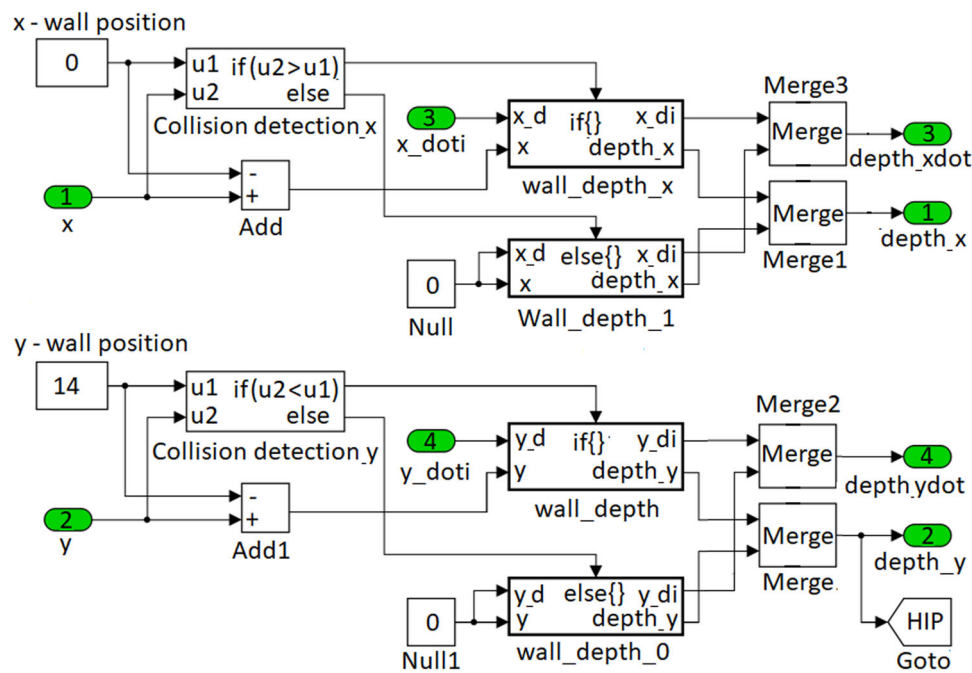


FIGURE 7 Implementation details of the *Collision Detection* module in Simulink/MATLAB.

human grasp impedance $H(s)$ (see Figure 1), which is subject-dependent and varies with grasp type, duration, and so on, the simulations were executed by allowing the end-effector to fall freely under gravity. The device's end-effector thus descended under its self-weight (a case of constant input impedance), until it encountered a horizontal virtual wall. In case any wall penetration was detected, a reaction torque was generated at the active joints in accordance with the two-DOF haptics force law described in Equation 13.

The end-effector's position and velocity responses resulted in damped oscillations for a stable output. In contrast, for unstable output, constant magnitude oscillations known as limit cycles were observed. Such oscillations typically exceeded a single encoder resolution in magnitude. To precisely characterize the simulation response as being stable or unstable, a stability criterion had to be defined. With this consideration, oscillations that exhibited a magnitude greater than one encoder resolution and that persisted beyond 2 s were classified as unstable. To identify the stability limits, we incrementally increased the virtual stiffness coefficient, K_v , while maintaining a specific virtual damping coefficient, B_v , until the aforementioned oscillatory behavior was observed. The value at which this behavior occurred was recorded as the stable virtual stiffness for that particular virtual damping coefficient. This process was repeated to generate the entire Z -width of the simulated two-DOF haptic interface.

6 | RESULTS AND DISCUSSION

In the preceding sections, the hardware implementation and the simulation strategy of the dual-rate sampling scheme for a custom-developed two-DOF haptic interface have been detailed. This

multi-DOF extension of the dual-rate control was attempted in anticipation of gaining improvements in the *Z-width* of such multi-DOF interfaces. In this section, the experimental and simulation-based results are presented and discussed in some detail.

6.1 | Experimental results

The experimental protocol involved the user grasping the device end-effector and exerting a motive force directed at the virtual wall, as illustrated in Figure 8A. The operator underwent training before the experiments to strike the wall near a specific position repeatedly and was also made aware of the criterion for instability—*sustained vibrations along with a loud sound*. In an alternate experimental protocol, shown in Figure 8B, the end-effector was allowed to fall freely without any input human impedance. Figure 9 showcases the typical wall hit profiles observed during the experimental trials.

During the experiments, the position loop was sampled at four distinct frequencies and the maximum sampling frequency was restricted to 20 kHz. Although much higher sampling frequencies could also be used, they generally require the concurrent use of higher-resolution encoders.¹¹ Furthermore, higher sampling rates are also limited by the motor saturation considerations. The sampling rate of the velocity loop was, however, maintained constant at 2 kHz for all the experimental trials to avoid noisy velocity estimates. To evaluate the Z -width, the controller maintained a value of zero for f_x , considering that the virtual wall was implemented solely along the Y -direction. For a specific combination of sampling rates, the value of K_y was incremented in fixed steps, for a selected value of B_y in the FPGA. At the onset of instability, the previous value of K_y was noted to be the maximum

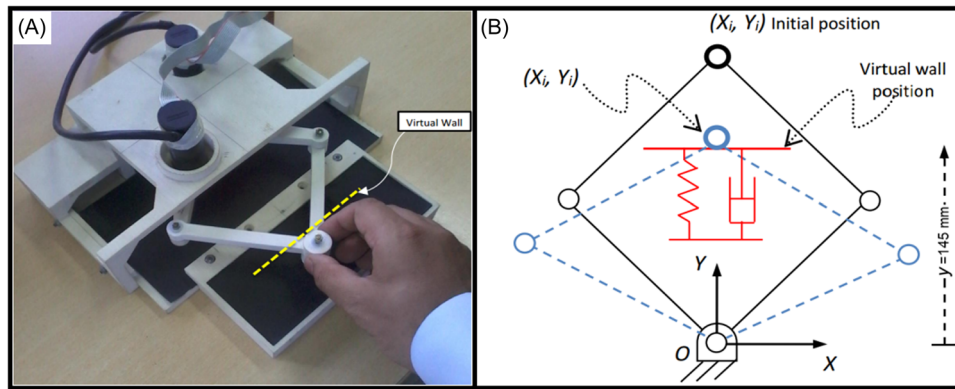


FIGURE 8 Depiction of the two degrees-of-freedom experimental protocols. (A) Human operator in the loop; (B) gravity-based test.

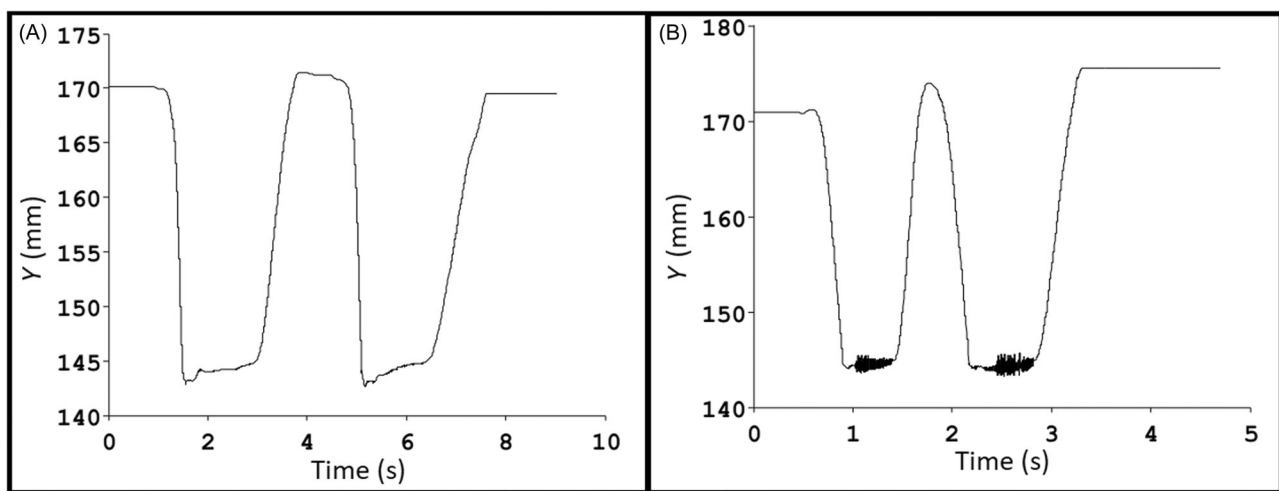


FIGURE 9 Two degrees-of-freedom wall hit profiles during a selected experimental trial with the human operator in the loop (wall position at $y = 145 \text{ mm}$). (A) Stable interaction with the virtual wall during the human operator in the loop; (B) unstable interaction with the virtual wall during the human operator in the loop.

renderable stiffness at that particular damping value and the frequency combination. The experimental trials were repeated multiple times, with different combinations of K_y and B_y coefficients to obtain the entire Z-width. The same is depicted in Figure 10.

In the case of the uniform-rate sampling scheme, specifically the (2–2) frequency combination, the virtual stiffness, K_y , increased from 295 to 440 N/mm when the virtual damping, B_y , was increased from 0 to 0.5 Ns/mm. Furthermore, by increasing the value of B_y to 2.2 Ns/mm, K_y further increased to 505 N/mm. However, when B_y was increased to 5.0 Ns/mm, K_y decreased to 150 N/mm, thereby establishing the maximum range of virtual damping for the (2–2) scheme. With the (20–20) sampling frequency combination, the presence of virtual damping did not contribute to an increase in virtual stiffness due to noisy velocity estimation at 20 kHz. Figure 11 illustrates a comparison of the active joint velocities obtained during a set of automated experiments using 20 and 2 kHz sampled position data. During these automated experiments, a constant torque was applied to the motors controlling the active joints to simulate contact with a virtual wall at a

specific velocity. Upon encountering the virtual wall, the end-effector penetrated the wall till a stable contact was established. The resulting joint velocities and the end-effector velocity components were recorded. Figure 11 clearly demonstrates that the velocity in the (20–20) case exhibited significantly more noise when compared to the (20–2) case for both active joints, joint 1 and joint 3, of the haptic interface. The same was witnessed in the end-effector velocity components as well. This poor velocity estimation at the sampling frequency of 20 kHz resulted in instability of the virtual wall and, in certain instances, led to an unsuccessful implementation of virtual damping, ultimately reducing the interface's Z-width.

By employing the dual-rate sampling scheme, the two-DOF haptic interface demonstrated improved performance, allowing for the preservation of the implementation range of virtual damping even when the sampling frequency of the position loop was increased to 20 kHz. This fact becomes evident when examining Figure 10, where it can be observed that the maximum value of virtual damping remained around 5.0 Ns/mm, while the maximum stiffness reached a

peak of 700 N/mm. This improvement was primarily attributed to the better velocity estimation using 2 kHz sampled data in the dual-rate scheme. For all the frequency combinations considered here, the peak renderable stiffness increased as the position loop sampling frequency increased. This observation is consistent with the established results available in the literature.^{11,20,22} The stable implementation range of virtual damping was equal for all the cases owing to the use of the same velocity loop sampling frequency. Similar results were obtained for the one-DOF implementation of the dual-rate control scheme reported in our earlier work.⁴⁴ These results establish the successful application and the advantages gained by employing the dual-rate control scheme for the implementation of multi-DOF haptic devices.

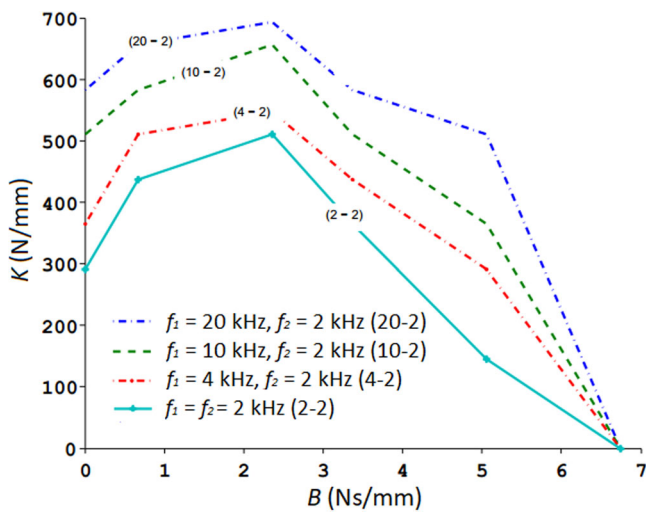


FIGURE 10 Experimental Z-width plots of the two degrees-of-freedom (DOFs) haptic interface with the dual-rate sampling scheme, demonstrating the advantages of having independent sampling rates for the multi-DOF haptics force law implementation.

6.2 | Simulation results

During the simulations, the horizontal component of the position and velocity was considered negligible due to the fact that the motion of the end-effector was purely vertical, and the virtual wall was defined as a constraint along the Y axis. Besides this, the wall interactions were characterized as stable or unstable based on the previously defined stability criterion in Section 5.3. In Figure 12A, we can observe the simulated position profile during a stable interaction with the virtual wall. The figure shows that the end-effector's position dropped from its initial position of $y_i = 175$ mm until it reached the virtual wall at $y_w = 145$ mm. Upon hitting the virtual wall, the position exhibited oscillations for approximately 800 ms before gradually ceasing completely. Figure 12B illustrates the position profile for an unstable virtual wall interaction, characterized by persistent oscillations. These oscillations, known as limit cycles, remained constant in magnitude and persisted even after a simulation duration of 2 s, indicating the presence of instability.

To facilitate a thorough comparison with the experimental observations made earlier, repeated simulations with varying B_y and K_y values were carried out to determine the Z-width of the two-DOF device for several sampling rate combinations. Figure 13A presents the simulation results, which exhibit a similar trend to the experimental findings shown in Figure 10. However, it is worth noting that the simulation results were more conservative compared to the experimental results. This is primarily due to the omission of several nonlinear factors like actuator saturation, amplifier bandwidth, structural compliance, and so on, during the simulations that otherwise creep in during the actual experiments. Additionally, this is also attributed to the noninclusion of human grasp impedance during the simulations. For convenience and clarity, a comparison of experimental and simulated Z-width plots is shown for a specific combination of sampling rates in Figure 13B. As is evident, the simulation results are in good agreement with the experimental outcomes, and likewise, the implementation range of virtual damping is preserved even after increasing the frequency of the position loop.

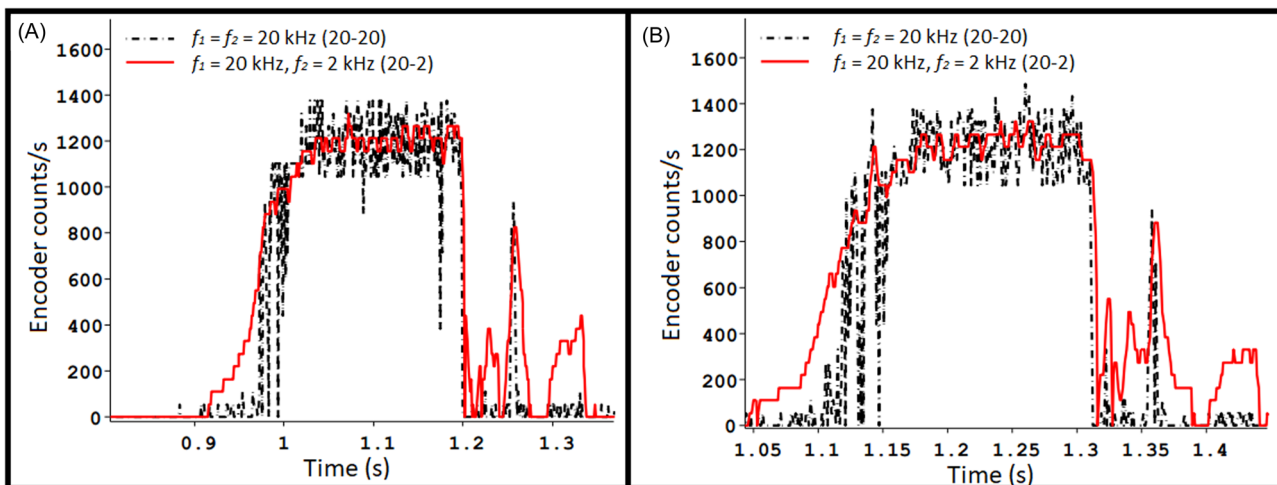


FIGURE 11 Comparison of the active joint velocity profiles of the haptic device during automated wall hit trials (experimental). (A) Comparative velocity profile of one active joint (1) of the device; (B) comparative velocity profile of the other active joint (2) of the device.

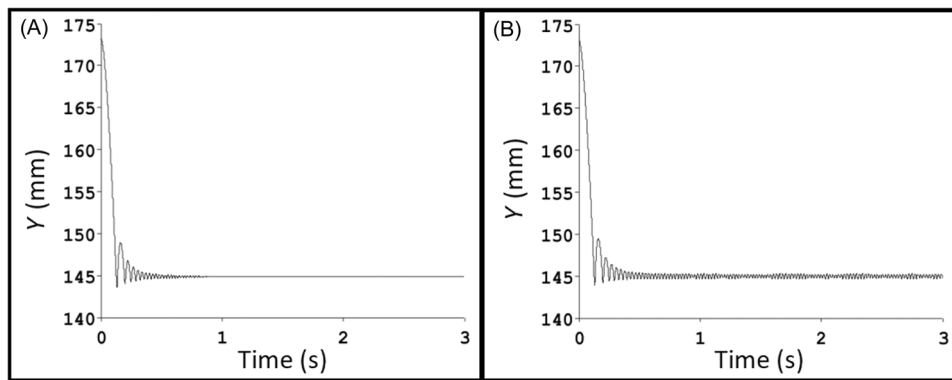


FIGURE 12 Position profiles during stable and unstable hits of the two degrees-of-freedom device with the virtual wall during a particular simulation. Note that the initial end-effector position is specified as $y_i = 175$ mm and the wall location is $y_w = 145$ mm. (A) Stable interaction with damped oscillations—simulation; (B) unstable interaction with continuous oscillations—simulation.

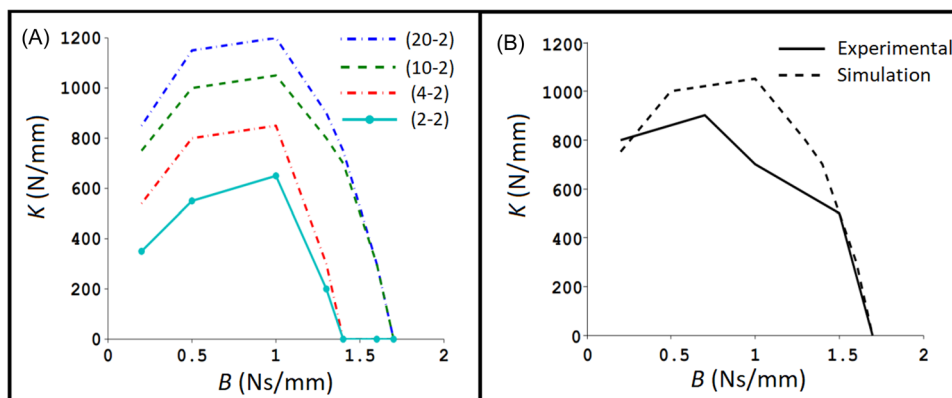


FIGURE 13 Simulated Z-width plots of the two degrees-of-freedom (DOF) haptic interface with the dual-rate sampling scheme. (A) Z-width plots (K_y vs. B_y) for the two-DOF haptic interface for various frequency combinations; (B) comparison of experimental and simulated Z-widths for 10-2 frequency combination.

7 | CONCLUSIONS

In conclusion, this study explored the promising potential of the dual-rate sampling scheme to enhance the performance of haptic interfaces, particularly in the context of multi-DOF systems. The innovative approach of utilizing two distinct sampling rates effectively addressed the challenge of reduced Z-width, offering greater stability, even at higher controller sampling rates, and thereby setting it apart from the conventional uniform-rate control scheme. This represents a significant advancement in the field of haptic technology.

A central focus of our research involved the intricate implementation of the dual-rate sampling scheme on an FPGA at the digital level. Despite the complex nature of this task, we successfully developed digital logic for crucial components such as velocity estimation filters, collision detection mechanisms, and the incorporation of device kinematics. Our in-house developed two-DOF pantograph served as the haptic interface, and the FPGA emerged as a powerful platform for executing the controller strategy.

In summary, our research showcases the potential of dual-rate sampling in the realm of haptic technology, with practical applications in

multi-DOF systems. The integration of FPGA-based implementation with digital logic for velocity estimation and collision detection further extends its usability. This work contributes valuable insights to the development of more stable and high-performance haptic interfaces, offering a promising avenue for future advancements in the field.

Our further research will delve into optimizing the hardware implementation on FPGA with regard to memory usage, resource allocation, timing limitations, and other pertinent considerations.

CONFLICT OF INTEREST STATEMENT

The authors declare no conflict of interest.

DATA AVAILABILITY STATEMENT

Data sharing are not applicable to this article as no new data were created or analyzed in this study.

ORCID

Majid Koul  <http://orcid.org/0000-0003-0039-4778>

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AUTHOR BIOGRAPHIES



Majid Koul completed his PhD degree in 2015 from the Indian Institute of Technology Delhi (IIT Delhi). During his research, he worked on impedance control of Haptic Devices, Multibody Dynamics, Robotics, and Mechatronics applications. Majid Koul graduated in Mechanical

Engineering from the National Institute of Technology Srinagar (NIT Srinagar) in 2006. Currently, he is working as a senior assistant professor in the Department of Mechanical Engineering at the Islamic University of Science and Technology (IUST) Kashmir, J&K, India. His research interests span Mechatronics, Robotics, Haptics, Multibody Dynamics, and their applications for education, assistance, and rehabilitation.



Suhail Khosa completed his PhD degree in 2023 from the National Institute of Technology Srinagar, J&K, India, with a specialization in Haptics, Robotics, and Control Engineering. His doctoral research focused on the stability and reduced-order modeling of dual-rate haptics controllers. Suhail graduated

in Mechanical Engineering from the University of Kashmir in 2015. Currently, he is working as an assistant professor (tenure) in the Department of Mechanical Engineering at the National Institute of Technology, Srinagar, J&K, India. Prior to this, he worked as a lecturer (tenure) at the Institute of Technology, Zakura, Srinagar, and SKUAST Shalimar, Srinagar, J&K, India. Suhail's current research focus is on the design and development of deformable haptic devices for low-impedance rendering. He is also interested in the haptic rendering of fluid media for virtual reality-based applications and the dynamics formulation of haptic and robotic devices using machine learning-based techniques.



Babar Ahmad received the ME and PhD degrees in Mechanical Engineering from the Indian Institute of Science (IISc), Bangalore, India, in 1999 and 2012, respectively. He is currently working as a professor in the Department of Mechanical Engineering, the National Institute of Technology, Srinagar (NIT Srinagar). During his research at the Indian Institute of Science, Bangalore, he worked on the design and development of a MEMS (Micro Electro Mechanical Systems) ultrasonic transducer for NDE applications. His current research interests include MEMS ultrasonic transducers, Dynamics of MEMS devices, Haptics and Control, and Vibroacoustics.

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